YUE FAN

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EDUCATION

Case Western Reserve University, Cleveland, OH *Aug 2022 – Present*
Ph.D. Candidate, Systems and Control Engineering (Expected May 2027)
- Advisor: Behnam Malakooti, Ph.D.
- Dissertation: Geometric Dispersion Theory for Decision Applications (tentative)

Oregon State University, Corvallis, OR *Mar 2021*
B.S., Electrical & Computer Engineering; Minor in Computer Science

RESEARCH INTERESTS

Multi-criteria decision making (MCDM); Geometric Dispersion Theory (GDT); decision analytics and utility/value prediction; interpretable machine learning and hybrid ML–decision frameworks; decision making under risk; optimization modeling with applications in energy, finance, and quality assessment.

TECHNICAL SKILLS

- Programming: Python (PyTorch, TensorFlow, scikit-learn, NumPy, pandas), MATLAB, C/C++
- Optimization & Decision Science: convex optimization; linear, network, dynamic, integer, and nonlinear programming; MCDM (GDT, SAW, TOPSIS, VIKOR, AHP, MAUT, PROMETHEE); out-of-sample validation; experimental design
- Machine Learning & Data Science: regression, classification, clustering, neural networks, SVM, feature selection, dimensionality reduction; distributed training (PyTorch Lightning, MPI)
- Tools: Tableau, Excel Solver, LaTeX, Git; PCB design (KiCad)

RESEARCH IN PROGRESS

[1] GDT-FANN: A Geometric Dispersion Feedforward Network for Learning Multi-Criteria Evaluations
(Job Market Paper; preparing for submission to Management Science)
***Yue Fan**, Behnam Malakooti, Huanshuo Wang*
- Accepted for presentation at INFORMS Annual Meeting 2026, San Francisco

[2] The Predictive Value of GDT Aggregation in Data-Limited Multi-Criteria Value Models *(Preparing for submission to European Journal of Operational Research)*
***Yue Fan**, Huanshuo Wang, Behnam Malakooti*

[3] A Comparative Analysis of Multi-Criteria and Geometric Dispersion Theory Methods Using Out-of-Sample Validation with Application to Wind Farm Location Selection *(In revision for resubmission)*
*Behnam Malakooti, Abdulaziz Altowijri, Huanshuo Wang, **Yue Fan***

[4] A Geometric Dispersion Distribution for Separating Good and Bad Volatility in Stock Returns
(Working Paper)
***Yue Fan**, Huanshuo Wang, Behnam Malakooti*

[5] Analysis of Portfolio Selection Based on Geometric Dispersion Theory Risk Model *(Working Paper)*
*Mohamed Komaki, Behnam Malakooti, **Yue Fan**, Huanshuo Wang*

[6] Geometric Dispersion Theory under Multi-Criteria Decision Making and Out-of-Sample Experiments Based on Real-World Data *(Working Paper)*
*Huanshuo Wang, Behnam Malakooti, **Yue Fan***

PUBLICATIONS AND PATENTS

Fan, Y., & Wang, J. (2022). Solar cells fabrication procedures. In *Proceedings of the 3rd International Conference on Electronic Science and Automation Control (ESAC 2022)*. Bangkok, Thailand: Clausius Scientific Press. https://doi.org/10.23977/esac2022.004

Fan, Y. (2021). Modular pet feeding device. Chinese Utility Model Patent No. ZL 2021 2 2413328.4 (filed October 2021; granted).

RESEARCH EXPERIENCE

Network Lab, Case Western Reserve University *Sep 2023 – Present*
Graduate Research Assistant (Advisor: Behnam Malakooti, Ph.D.)

- Established the mathematical foundations of Geometric Dispersion Theory (GDT), proving that ranking-only MCDM methods (TOPSIS, VIKOR) cannot recover absolute utility values because their scores shift with the alternative set, while GDT produces context-independent utilities
- Developed GDT-FANN, a feedforward network that learns indirectly observed criteria in its hidden layer and aggregates them through a GDT readout, retaining interpretable criterion weights while achieving competitive out-of-sample accuracy; validated on wind-farm siting, Amazon product ratings, and health-insurance claims.
- Benchmarked interpretable GDT aggregation against traditional MCDM and predictive models for accuracy, stability, and interpretability under criterion and training-sample reduction (red-wine quality, social-worker decisions, and housing-price data).

Semiconductor Device Physics and Fabrication Seminar *Oct 2021 – Dec 2021*
Research Team Lead (Supervisor: Munib Wober, Ph.D.)

- Led a 5-person team evaluating wafer-cleaning and doping sequences for solar-cell efficiency; designed a P+/N-type cell structure that became the basis of the team's ESAC 2022 conference paper.

TEACHING EXPERIENCE

Teaching Assistant with Lecturing and Course Development, Case Western Reserve University

ECSE 360/460 Manufacturing and Automated Systems: Design and Optimization *Fall 2024, 2025, 2026*

- Co-designed the Multi-Criteria Decision-Making module and delivered lectures

ECSE 352 Engineering Economics and Decision Analysis *Fall 2024, 2025, 2026*

- Designed the Risk Decision module (mean–variance analysis, decision trees, expected utility theory, GDT) and taught multiple lectures

ECSE 350/450 Operations and Systems Design *Spring 2025, Spring 2026*

- Led Excel Solver instruction and exam review sessions; graded case-study projects

Teaching Assistant, Case Western Reserve University

ECSE 416 Convex Optimization in Engineering *Spring 2024, Spring 2025*
ECSE 346 Engineering Optimization *Spring 2024, Spring 2025*
EPOM 405 / ECSE 411 Applied Engineering Statistics (online graduate course) *Spring 2026*
ECSE 408 Introduction to Linear Systems *Fall 2023*
BUAI 445 Advanced Marketing Analytics (Weatherhead School) *Spring 2023*

PROFESSIONAL EXPERIENCE

xLab, Case Western Reserve University — Nestlé Purina Project *Spring 2023*
Student Consultant

- Advised Nestlé Purina, as part of a student consulting team, on the feasibility of a pet-data platform that would use in-home IoT devices to monitor pets and turn the resulting data into data-driven pet-insurance products.
- Proposed how IoT data could be captured and structured for downstream analytics, and outlined data-analysis and modeling approaches for pet-health prediction and insurance risk assessment; presented recommendations to the sponsor.

Olababy, Beaverton, OR *Jun 2021 – Mar 2022*
Product Development Manager

- Led development of an automatic baby-formula dispenser from technical requirements and feasibility analysis through PCB design and an engineering validation prototype.
- Translated brand and market requirements into technical specifications as liaison among design, engineering, and marketing teams; supported supply-chain operations.

SELECTED TECHNICAL PROJECTS

Autonomous Vision-Guided Quadcopter for Moving-Target Tracking *Fall 2026 – Present*

- Co-Investigator on a team project funded by a $2,500 Sears think[box] Student Project Fund grant (Project Lead: Ken Chen); the quadcopter autonomously tracks a moving ground vehicle and lands on it.
- Responsible for the vision-based tracking of the moving vehicle and the system-level decision making that coordinates tracking, approach, and autonomous landing.

Distributed Model Training with PyTorch Lightning and MPI *Fall 2023*

- Implemented distributed data-parallel training across multiple GPUs (BERT sentiment classification on IMDb); fine-tuned GPT-2 under constrained compute with real-time resource monitoring and early stopping.

PET Image Reconstruction Using Convex Optimization *Spring 2024*

- Formulated PET reconstruction as log-likelihood maximization; implemented gradient-ascent and quasi-Newton methods exploiting sparsity (1,080 voxels, 2,164 detector pairs); analyzed MLE vs. penalized-likelihood trade-offs.

Advanced Statistical Learning Implementations *Fall 2024*

- Built custom implementations of clustering, LAD/percentile regression, LASSO/ridge shrinkage, SVM, kNN, and multidimensional scaling; benchmarked against standard libraries.

SERVICE AND PROFESSIONAL AFFILIATIONS

- Peer Reviewer, INFORMS, 2026
- IEEE Student Member, since 2024

REFERENCES

Behnam Malakooti, Ph.D.
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